

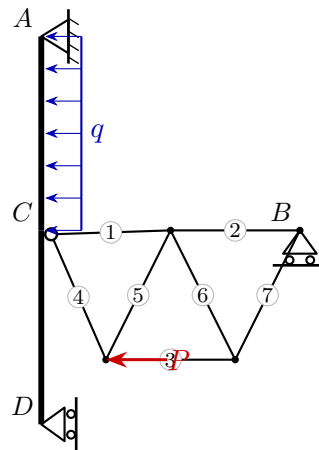
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VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

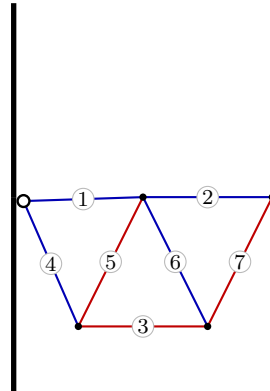
$$A: R_x = 14, R_y = -5 \quad D: R_x = 8, R_y = 0 \quad B: R_x = 0, R_y = 5$$

$$C: H_x = -10, H_y = 5$$

3. Truss member forces

Member	Force S [kN]	Type
1	-7.5	compression
2	-2.5	compression
3	5	tension
4	-5.59	compression
5	5.59	tension
6	-5.59	compression
7	5.59	tension

red: tension (+) blue: compression (–)



4. Section-force diagrams of the beams

